FACULTY OF COMPUTING

SEMESTER 1 2024/2025

INDIVIDUAL ASSIGNMENT

SECP3273 – SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 01

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MATRIC NO.: A23CS0252

TABLE OF CONTENT

1.0 Introduction

The system that I have proposed is the Course Registration System which allows the students to enroll the courses via online. The system functions to ease the course registration process by streamlining the available courses with the schedules of the faculty. The system provides a user-friendly interface for the students to browse the available courses according to their semester and secure their place in the course. The system is not only for the students but it also provides some functionality for both lecturers and administrators of the faculty. This makes the course registration process more transparent and efficient to the students.

1.1 Project Overview

The project aims to develop a secure and user-friendly system which makes the course registration process more flexible to the users which are students, lecturers and administrators. The users are able to manage their course registration and make changes to their course via the system.

1.2 Key Features

1.2.1 Add Courses

Students can add courses according to their current semester and see the details of the courses they have registered.

1.2.2 Changes Confirmation

This feature allows the user to rethink their decision to make changes to the system such as modify the registration and cancel the registration. This ensures the changes made by the user are on purposely not by accident.

1.2.3 Secure Login

This feature provides the user with a secure login in which they can enter their id and password to the system for credentials purposes. The password that will be stored in the database should be encrypted.

1.3 Significance of the System

The Course Registration System is a modernised system that transforms the traditional paper-based registration to the web-based registration process. This ensures the security of the data stored by having the user authentication process and enhances the accountability for the user to make changes of the registration.

1.4 Document Structure

This document is structured to provide a deep understanding to the system by providing the system design, their functionalities and the system implementation. Each section will have a detailed explanation of the specific aspects including their database design, development steps, and system interface.

Therefore, the system aims to simplify and streamline the course registration process for the user especially in university. Throughout the documentation, there will be more details on the system components and the system features.

2.0 System Design

This section provides both use case and entity relationship diagrams (ERD) for more clearer information of the system. It shows the flow of how the Course Registration System works and who is the person that is involved in the system.

2.1 Use Case Diagram

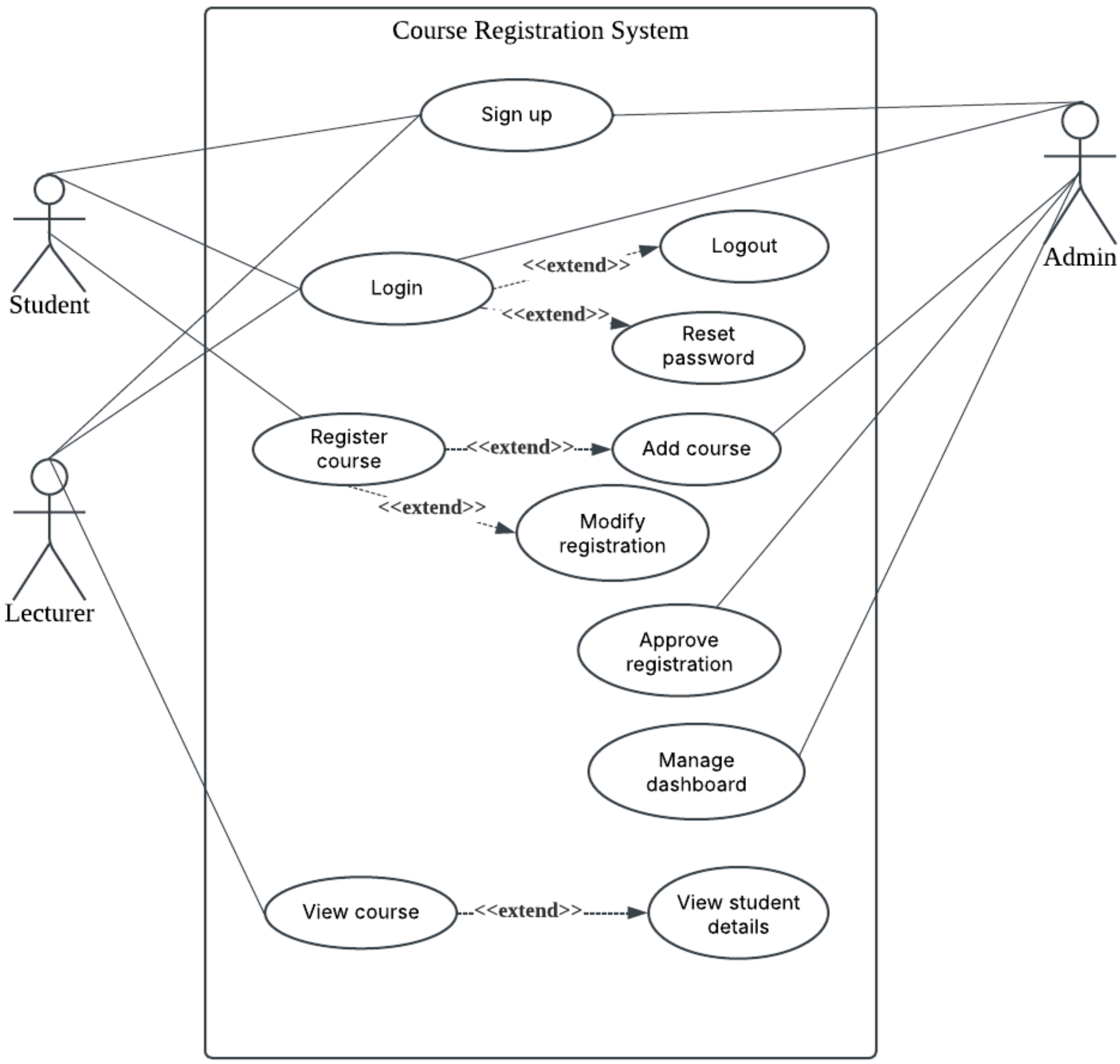

Figure 2.1.1 Use Case of Course Registration System

2.2 Entity Relationship Diagram (ERD)

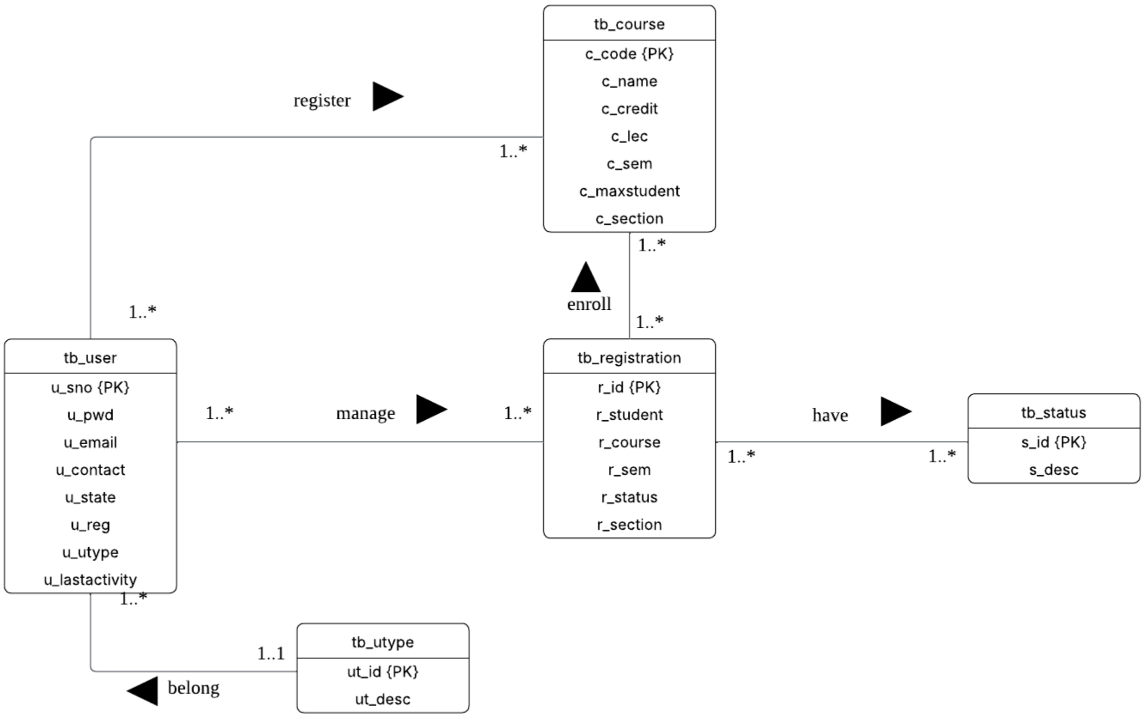

Figure 2.2.1 ERD Diagram of Course Registration System

3.0 Development Software, Language, Technology and Tools

3.1 Software

I have implemented using various software tools for this Course Registration System. Visual Studio Code is frequently used as I need the tools to code for the system and it also supports many kinds of programming languages. In addition, XAMPP has also been used for this project as it is an open-source universal web server solution stack package that enables the system to be hosted locally and offline by using the Apache HTTP server. Last, to be able to see how the system works from the code, I have used the Google Chrome browser to ensure all the functionality works seamlessly.

3.2 Language

The Course Registration System is handled with multiple programming languages to deliver its functionality. This includes Hypertext Preprocessor(PHP), Hypertext Markup Language (HTML), JavaScript, Cascading Style Sheets (CSS) and Structured Query Language(SQL). PHP stands for the server-side scripting language that serves as the connection between the front-end which is HTML and the back-end database. The HTML is meant to put structure to the web pages, while the CSS beautifies the content for an appealing presentation. JavaScript makes the system more interactive by responding to events generated by the user. Meanwhile, for MySQL which uses phpMyAdmin to store and retrieve the data, more specifically using SQL queries to perform necessary database operations for seamless data management.

3.3 Tools

For designing the Use Case and ERD diagram in this project, I have implemented the LucidChart which is an online platform that is specifically designed for creating visual diagrams as it contains many shapes that is easier for me to develop the ERD and Use Case diagram. This diagram is important as it is to illustrate the structure and user interactions of the system.

4.0 Database Design

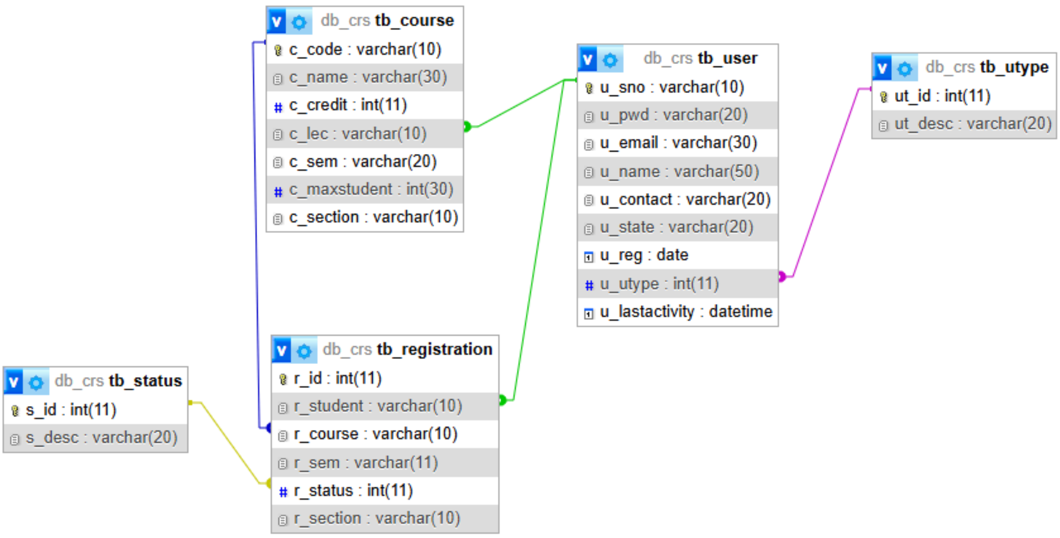

Figure 4.1 Relationship between tables of database

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	c_code	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
2	c_name	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
3	c_credit	int(11)			No	*None*			Change Drop More
4	c_lec	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
5	c_sem	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
6	c_maxstudent	int(30)			No	*None*			Change Drop More
7	c_section	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More

Figure 4.2 Structure of table "tb_course"

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	r_id	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	r_student	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
3	r_course	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
4	r_sem	varchar(11)	utf8mb4_general_ci		No	*None*			Change Drop More
5	r_status	int(11)			No	*None*			Change Drop More
6	r_section	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More

Figure 4.3 Structure of table "tb_registration"

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	s_id	int(11)			No	*None*			Change Drop More
2	s_desc	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More

Figure 4.4 Structure of table "tb_status"

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	u_sno	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More
2	u_pwd	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
3	u_email	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
4	u_name	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
5	u_contact	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
6	u_state	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
7	u_reg	date			Yes	*NULL*			Change Drop More
8	u_utype	int(11)			No	*None*			Change Drop More
9	u_lastactivity	datetime			Yes	*NULL*			Change Drop More

Figure 4.5 Structure of table "tb_user"

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**ut_id**	int(11)			No	*None*			Change Drop More
☐	2	**ut_desc**	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More

Figure 4.6 Structure of table "tb_utype"

5.0 Development Steps

To develop this system, I have implemented four steps of Software Development Life Cycle (SDLC) which contains planning and information gathering, design, development, and testing. First, for the planning and information gathering, I have examined the system's purpose, objective, target audience and scope from the lecturer's case study. This project is aimed to develop a web-based system which helps to simplify the course registration system process by providing the user with course lists, course register process and manage their course registrations. The target audience of this system are students, lecturer and administrator which are the people that usually will manage the course registration system.

Furthermore, for the design of the system, I have created the use case and ERD diagram to have clearer insights of the system development. A visually appealing layout was designed to the system and there are elements such as buttons and navigation bar to enhance the user experience to the system. The design phase needs to be given with special attention which involves producing wireframes that highlights the structure and layout of the system. This phase helps to ensure a solid and aesthetically pleasing base for the next stages of system implementations.

Next, for the development phase, since the lecturer already teaches in class on how to develop a basic structure for the system and I have dive into the system construction and some of the programming language, I came to a clear concept of the objective of the system. The development is conducted in a manner that assures a comprehensive approach. In determining the core functionality of the system, the security issues were also prioritized to the system. This includes implementing the password encryption, protection against the SQL injection attacks, as well as managing login errors. These measures reflect an agenda toward creating a secure, robust and user-friendly system.

Last, for the testing phase, I measure the system's performance using my own laptop where I frequently check and identify the errors that appear within the system. This action is to ensure the system works smoothly and has efficient operation. In addition, the user testing was conducted on the web server to gather valuable feedback and insights that can improve the system's usability. This interactive process and refinement of the testing process helps to deliver a reliable and user-friendly course registration system.

6.0 System Interface

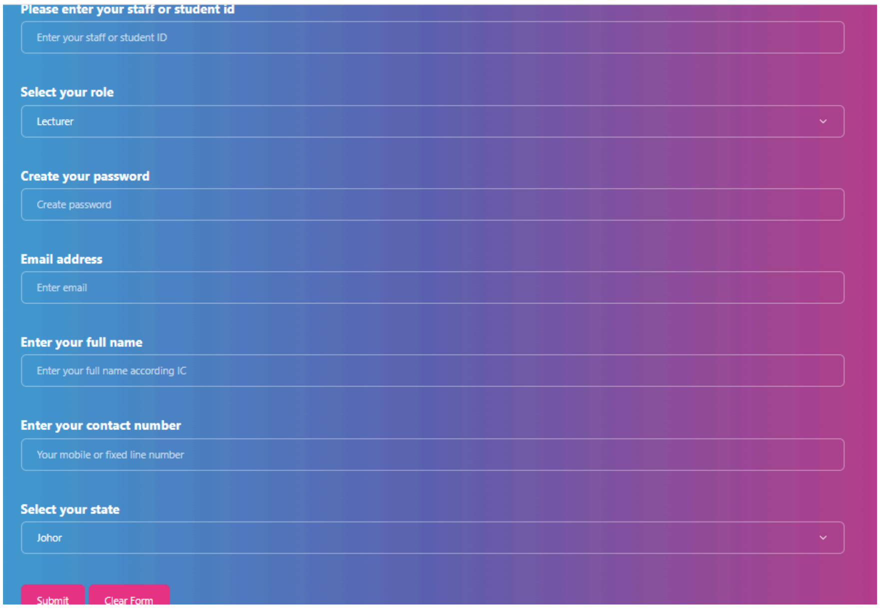

Figure 6.1 Sign up Page

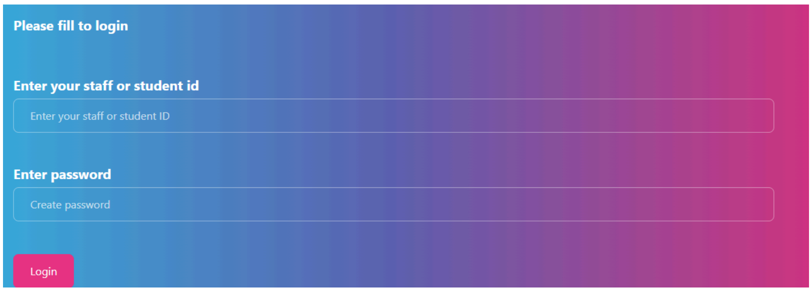

Figure 6.2 Login Page

Figure 6.4 Dashboard

Figure 6.5 Navigation bar

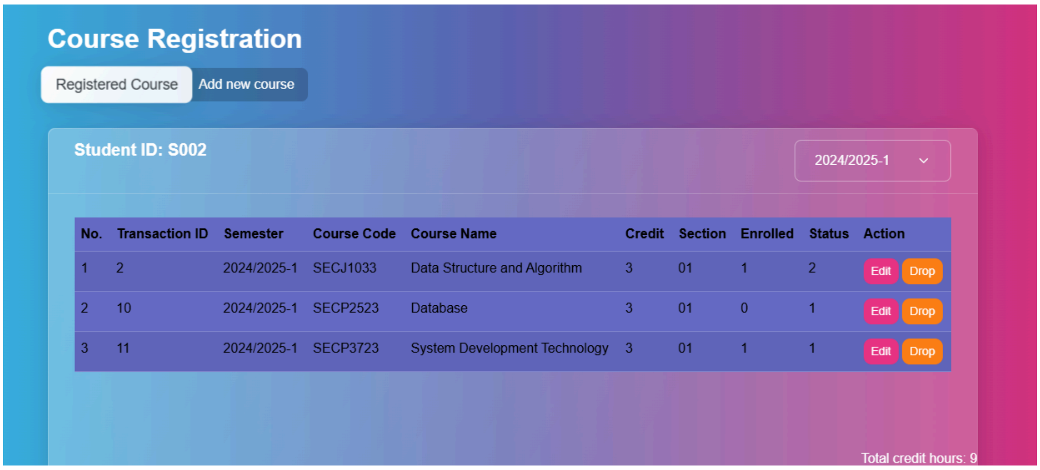

Figure 6.6 Registered Course page

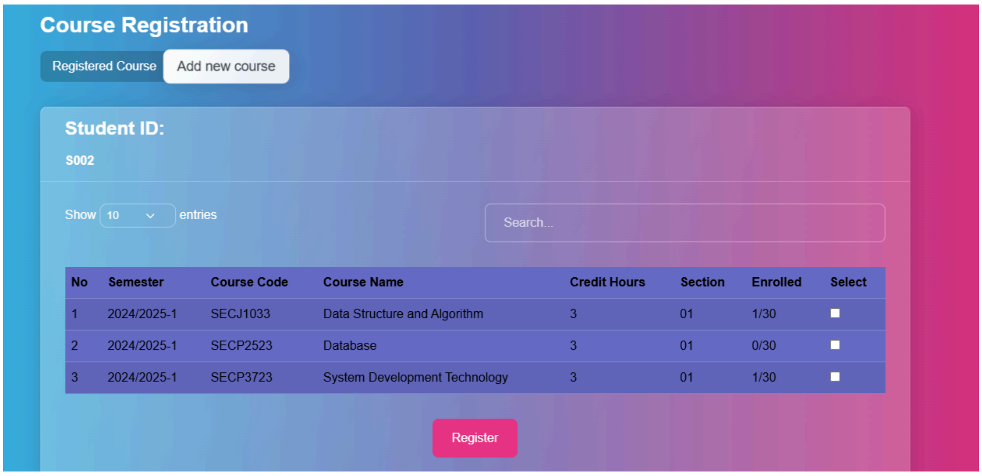

Figure 6.7 Add course page

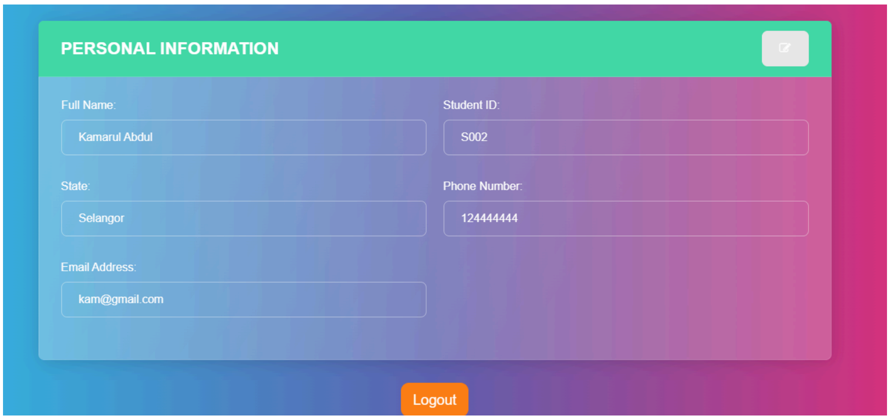

Figure 6.8 Student profile page

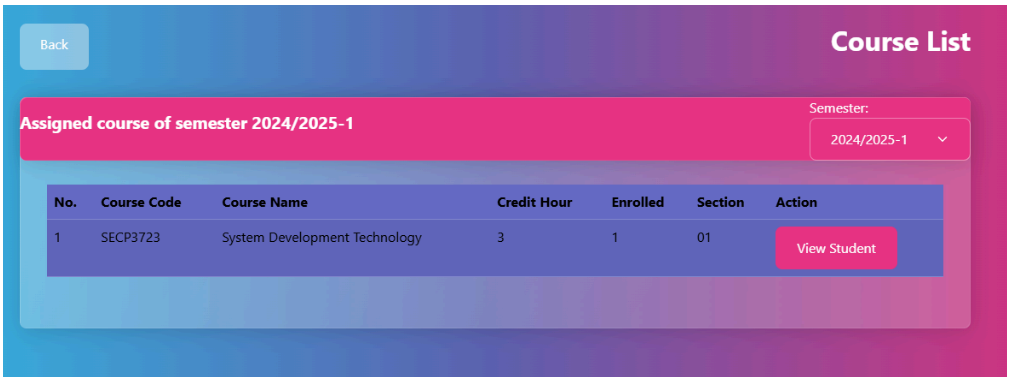

Figure 6.9 Lecturer course list page

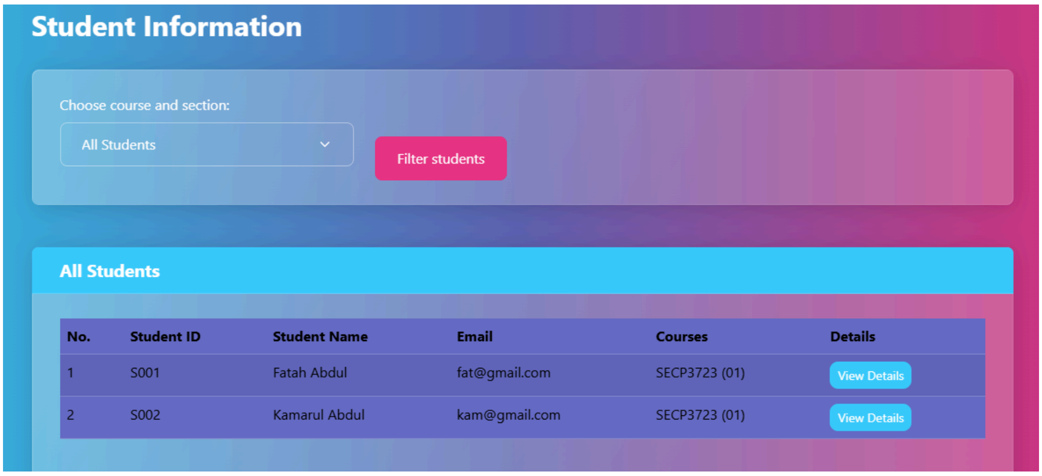

Figure 7.0 Lecturer view student information page

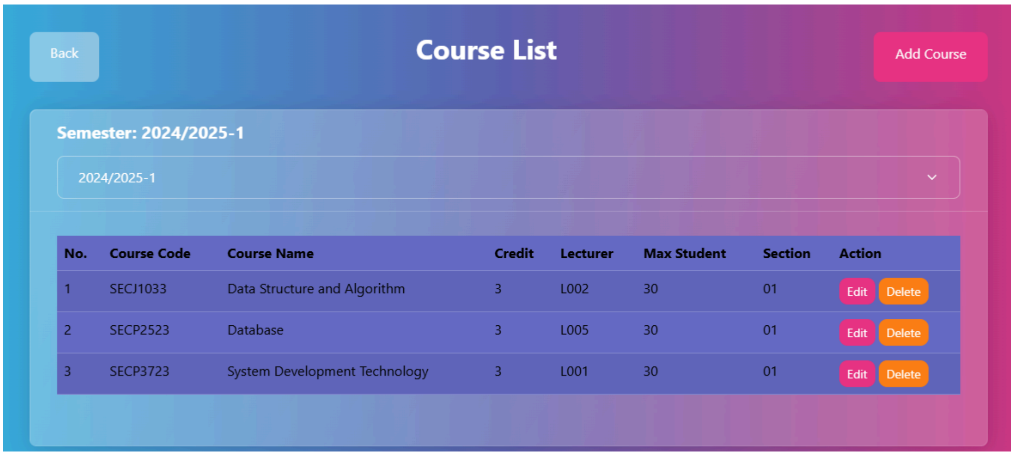

Figure 7.1 Admin Course list page

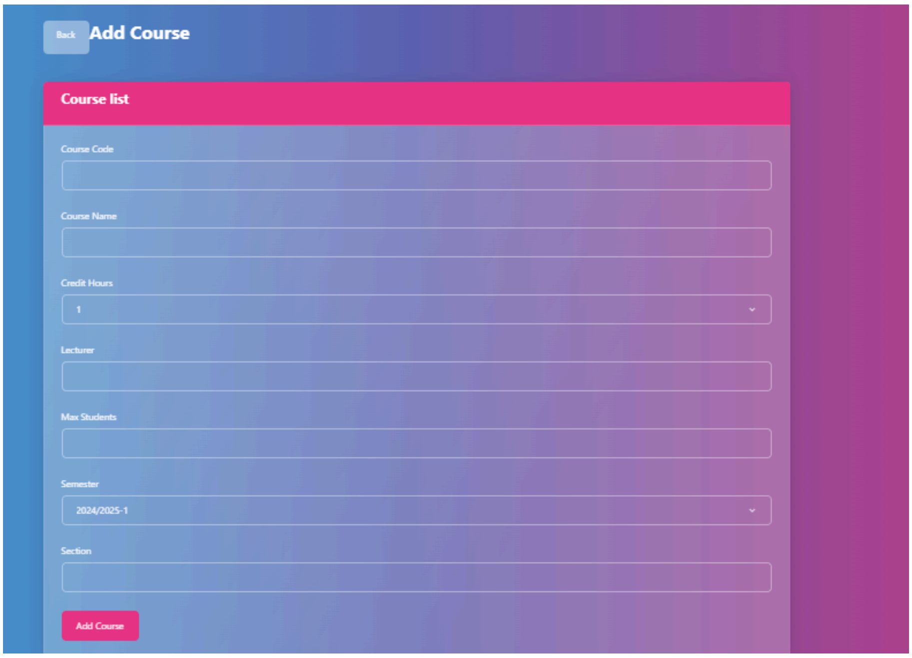

Figure 7.2 Admin add course page

7.0 Localhost Setup

1. Download and install the XAMPP Control Panel form https://www.apachefriends.org/download.html
2. Start Apache and MySQL.

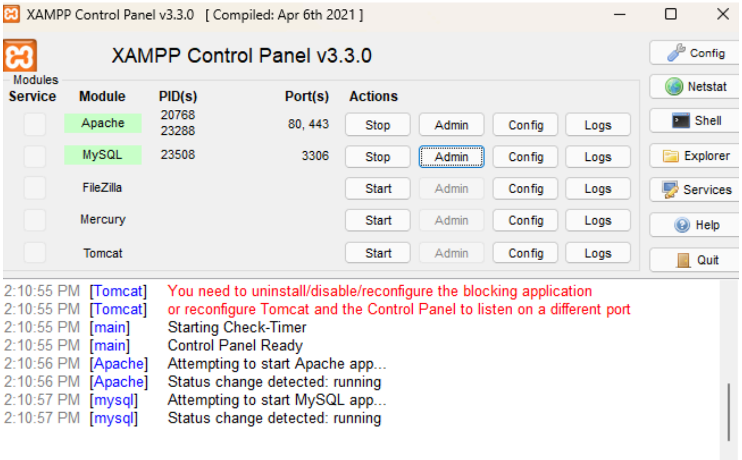

Figure 7.3 Xampp setup

3. View phpMyAdmin by clicking the ‘Admin’ button on the ‘Actions’ of the MySQL module.
4. Extract the ‘crs’ file into Local Disk (C:) > xampp> htdocs file path.

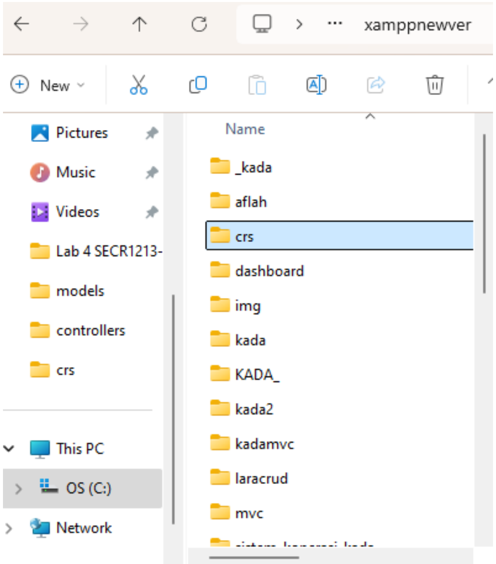

Figure 7.4 Crs file path

5. Create a new database with the name ‘db_crs’ using MySQL.

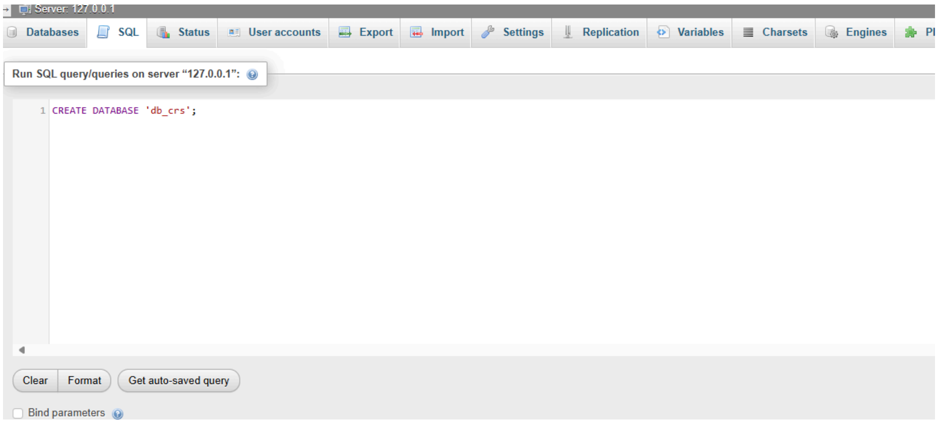

Figure 7.5 Create database

6. Import ‘db_crs.sql’ from the folder ‘crs’ to database ‘db_crs’.

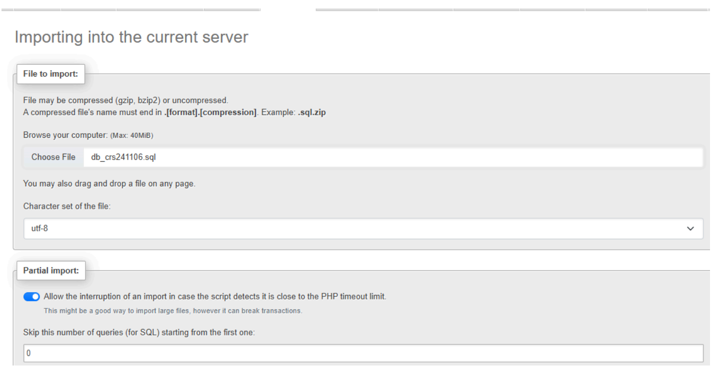

Figure 7.6 Import database

8.0 Hosting URL

http://localhost/crs/

9.0 User Credentials

1. Student ID: S001

   Password: 123

2. Lecturer ID: L001

Password: 123

3. Admin ID: A001

   Password: admin123